MOHIT MISRA

Full Stack GenAI Engineer · CTO & Technical Lead

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Portfolio: dev-mhtmsr.github.io | aroscai.com | knowledgespark.in

CORE SKILLS

GenAI / LLM: RAG (hybrid keyword + vector), embeddings, semantic search, chunking, prompt design, structured JSON outputs, LLM-vision extraction, Anthropic Claude API, LangChain

Agentic AI: LangGraph multi-agent workflows, model routing, tool / function calling, MCP (Model Context Protocol) servers, Langfuse tracing

Backend: Python (async), FastAPI, Pydantic, SQLAlchemy, Alembic, Node.js, Express.js, REST, SSE streaming, JWT, RBAC, Redis + arq workers

Frontend: React, Next.js, Vite, Tailwind CSS

Data: PostgreSQL, pgvector (HNSW), PgBouncer, row-level security, MongoDB

Cloud & DevOps: AWS (EC2, ECR, SSM, S3, SES, RDS), Docker, Docker Compose, GitHub Actions (OIDC), Nginx, Caddy, Cloudflare

Leadership: system architecture, technical roadmaps, team leadership (8 engineers), vendor management, production ownership, AI cost governance

IoT: ESP32 / ESP8266, MQTT, Alexa Skills, Google Assistant

EDUCATION

B.Tech, Computer Science & Engineering
National Institute of Science & Technology (NIST), BPUT
2017 – 2021 · CGPA 9.06

CERTIFICATION

RAG and Agentic AI Professional Certificate – IBM (verified)

PUBLICATIONS

Two-Wheeler Anti-Theft System Based on Three-Layer Architecture – Springer Singapore, 2022

RF-Based Social Distance Smart Band System – Apple Academic Press, 2022

ACHIEVEMENTS

Finalist (Top 10), CSI In-App Student Project Awards 2020

Top 9 Teams, NASA Space Apps Challenge 2019

Certificate of Excellence, NIST (SGPA 9.21)

SUMMARY

Co-founder, Director & CTO and full-stack GenAI engineer with 5+ years of building and running production software. Owns systems end to end: architecture, FastAPI / React / PostgreSQL backends and frontends, RAG chatbots, LLM-vision pipelines and agentic workflows, through AWS deployment, CI/CD and AI cost control. Led an 8-engineer team, managed an external mobile-app vendor, and shipped two live platforms used daily by customers, staff, learners and recruiters.

PROFESSIONAL EXPERIENCE

Co-Founder, Director & CTO – AROSCAI Technologies Pvt. Ltd. Nov 2021 – Present
Smart-home automation company · Berhampur, Odisha · aroscai.com

- Own technology strategy, architecture and delivery; led an 8-engineer team on the IoT smart-home platform and shipped Alexa and Google Assistant voice integrations in-house.
- Directed an outsourced partner (Hyperlink Infosystem) through the iOS / Android companion apps: scoping, milestone reviews and acceptance.
- Architected the production GenAI support chatbot (FastAPI, Claude, LangGraph): SSE token streaming, hybrid RAG (keyword / tag scoring + pgvector HNSW), a Haiku router over six category agents, and bounded tool calls through an MCP server for OTP-gated warranty lookups.
- Built the knowledge-ingestion pipeline (PDF / HTML extraction, paragraph-aware chunking, local CPU embeddings) with an admin tool so non-engineering staff update the knowledge base without code changes.
- Shipped AI Smart Estimate: switchboard photo → LLM-vision JSON → Pydantic validation against real panel sizes → catalogue match. One model call per photo; user corrections re-run only deterministic matching, keeping cost per estimate predictable.
- Built the Online Information System, the company's ERP / CRM (React, Vite, JWT, per-module RBAC, audit log): leads, tickets, customers, installations, warranties, partner hierarchies, and per-feature AI usage & billing reconciled with Anthropic billing.
- Run production on AWS with Docker Compose and Caddy; GitHub Actions CI/CD via OIDC → ECR → SSM (no stored AWS keys, no open SSH), health-check-gated releases, nightly database backups to S3.
- Built growth infrastructure: 19 programmatic SEO landing pages, JSON-LD, dynamic sitemap and OG images, GA4 and Meta Pixel conversion tracking.

Full Stack AI Developer – VLSIPROF Technologies Pvt. Ltd. Mar 2025 – Present
Knowledge Spark learning & hiring platform · knowledgespark.in

- Rebuilt Knowledge Spark from WordPress / Tutor LMS into a FastAPI + Next.js + PostgreSQL SaaS with four roles (student, instructor, admin, recruiter) on one account system, supporting multi-role users.
- Built courses, quizzes, certificates, Razorpay memberships, coupons, refunds and an instructor earnings ledger with payouts; email, payment webhooks and certificate PDFs run on Redis / arq background workers.
- Built recruiter semantic search (started as TalentTrack, a MERN recruitment app, later merged in): embeddings of full student profiles in pgvector, filtered by location, score and certifications, ranked globally before pagination.
- Scaled reads with nightly summary tables for admin / instructor dashboards and PgBouncer transaction pooling; enforced tenant data access with Postgres row-level security.
- Migrated WordPress users, courses, lessons and quiz attempts with a pilot run and checksum-verified files; deploy and maintain on a VPS with Docker Compose and Nginx.

Digital Design Engineer – Marquee Semiconductor Pvt. Ltd. Dec 2020 – Nov 2021

- Built ESP32 / ESP8266 IoT data-acquisition devices and backend pipelines streaming real-time sensor data to cloud servers.
- Started an internal project-management tool that later grew into corplink.ai, an independent startup.

PROJECTS (IN PROGRESS)

AgentDesk – Multi-agent support assistant: LangGraph state machine orchestrating retrieval, ticketing and escalation agents; internal tools exposed via MCP servers; pgvector retrieval with reranking.

DocMind – MCP-native document research agent: multi-step LangGraph reasoning with search, summary and citation tools usable by any MCP client; hybrid retrieval with structured JSON output.